Aditya Chauhan

Ludhiana, Punjab | +91 99888 77132 | aditya.chauhan.nitj.ac@gmail.com
adityac.codes | linkedin.com/in/aditya-chauhan-nitj | github.com/0xAditya-Labs

EDUCATION

Dr. B.R. Ambedkar National Institute of Technology Jalandhar — Aug 2023 – Jun 2027
Bachelor of Technology in Computer Science and Engineering — **CGPA: 8.56 / 10.0**

DAV Public School, BRS Nagar, Ludhiana — Jul 2022 – Jun 2023
Class XII, CBSE — **Percentage: 94%**

TECHNICAL SKILLS

- **Languages:** C++, Python, JavaScript, SQL, C
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks, Low-Level Design
- **Web Development:** React.js, Node.js, Express.js, FastAPI, REST APIs, WebSockets, MongoDB, MySQL, Tailwind CSS
- **Generative AI:** Retrieval-Augmented Generation (RAG), LangChain, LangGraph, ChromaDB, Sentence Transformers
- **Systems & Tools:** Multithreading, Concurrency, Redis, OpenTelemetry, Git, GitHub, Postman, Langfuse

EXPERIENCE

Accenture — Bengaluru
Advanced Application Engineering Intern — Jun 2026 – Jul 2026

- Built an internal LangGraph ReAct agent prototype with a FastAPI backend to benchmark retrieval performance.
- Optimized vector-search latency by **69%** (to 180ms) via OpenTelemetry and refactoring ChromaDB client to singleton pattern.
- Decreased LLM input-token usage by **34%** via doc-string tool-gating, suppressing **93%** of out-of-scope API calls.

Google Developer Group — NIT Jalandhar
Competitive Programming Lead — Dec 2024 – May 2026

- Mentored **70+** juniors in advanced DSA cohorts, boosting contest problem solve-rates by **40%**.
- Authored **35** original algorithmic problems with test cases and reference solutions across 16 campus contests.

PROJECTS

SwiftCache – Thread-Safe Concurrent LRU Cache Server — May 2026 – Jun 2026
C++17, TCP, POSIX Sockets, Concurrency, Multithreading — GitHub | Benchmark

- Architected a C++17 multithreaded TCP cache server featuring a custom binary/text wire protocol for network I/O.
- Sustained **160,000+ QPS** at $p99 <$ **0.24ms** by implementing a hybrid TTL eviction policy with active background reaping.
- Mitigated hot-key contention across **512** concurrent connections via 16-shard lock striping, achieving linear scaling on a **1M-key** Zipfian workload.
- Profiled median RTT (**87 µs**), where C++ cache logic consumes $<1\%$ ($\sim$300ns), while the Linux TCP stack accounts for $>99\%$.

Interlock – Distributed Real-Time Messaging Platform — Mar 2026 – May 2026
MERN Stack, Socket.io, Redis, JWT, Cloudinary, TailwindCSS — Live Demo | GitHub

- Engineered distributed WebSocket connections and private API routes using JWT authentication and Bcrypt hashing.
- Achieved **100% message delivery** at **35ms** $p95$ latency for **400 concurrent WebSockets** by architecting a dual-server backend with Redis Pub/Sub to bypass event loop saturation.
- Eliminated multi-tab false-offline race conditions by implementing atomic reference counting (`INCR/DECR`) in Redis.
- Minimized message payloads by **80%** by offloading media processing from base64 to an asynchronous Cloudinary CDN..

COMPETITIVE PROGRAMMING & ACHIEVEMENTS

Profiles: LeetCode | Codeforces | CodeChef | GeeksforGeeks

- **Meta Hacker Cup 2025:** Advanced to Round 2, securing Global Rank **3172** (AIR 875).
- **Flipkart GRiD 2025 and 2026:** Semi-Finalist in 7.0 (2025) and 8.0 (2026), placing in top **0.5%** among 170K+ participants.
- **Adobe University Hackathon 2026:** Shortlisted for Round 2 (Top 1.5%) from **301K+** global applicants.
- **LeetCode Knight (Top 1.5%):** Peak rating **1982**; Global Rank **315** in Biweekly Contest 172.
- **Competitive Profiles:** Specialist on Codeforces (Peak **1418**) | 3-Star on CodeChef (Peak **1661**).
- **Algorithmic Proficiency:** Solved **1,500+** complex algorithmic problems across various platforms.